

Lab 1 Report

Environment and First Notebook

A first look at the Palmer Penguins dataset in Google Colab

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1. Objective

This first lab had a technical goal and a habit-forming one: get a working Python environment in Google Colab and carry one dataset through the complete first-look workflow - load a CSV straight from a URL into a pandas DataFrame, report its shape, column types and missing values, select columns and filter rows, and draw one labelled figure - while starting the course routine of writing a short reading under every output and saving the work to GitHub. My own test for the session: after the inspection commands alone, I should be able to describe the dataset and its weak spots.

2. Data Description

The dataset is Palmer Penguins: physical measurements of 344 penguins of three species (Adelie, Chinstrap and Gentoo) collected on three Antarctic islands by Dr. Kristen Gorman's team. I used the standard CSV copy maintained in the seaborn-data repository on GitHub, loaded directly from its URL, so the notebook depends on no downloaded file. The columns come in two kinds, categorical and numeric, as Table 1 details.

I changed nothing this week; the work was inspection only, and the data is slightly messy: sex is missing in 11 rows and each measurement column in 2. Rather than stopping at the totals, I listed every incomplete row with the boolean filter `isna().any(axis=1)`. The eleven affected rows are not spread evenly - six are Adelie (five on Torgersen, one on Dream) and five are Gentoo, all on Biscoe - and nine of them lack only the sex, while rows 3 and 339 carry no measurements at all. Table 1 summarises the dataset and what was done to it.

Property	Value
Source	github.com/mwaskom/seaborn-data - penguins.csv, read from the raw URL
Size	344 rows $\times$ 7 columns
Target variable	none (descriptive lab); species used for grouping and colouring
Key features	numeric: bill length/depth, flipper length, body mass; categorical: species, island, sex
Missing values	sex: 11 rows; each numeric column: 2 - rows listed in the notebook, none dropped
After preprocessing	no cleaning applied; all 344 rows retained

Table 1. Dataset summary and preprocessing outcome.

3. Methodology

All work lives in a single Colab notebook that runs top to bottom after a kernel restart, using pandas 2.2.3 and NumPy 2.1.3 for the data and matplotlib with seaborn (whitegrid style) for plotting. I worked through the tasks in this order:

1. **Load and peek.** `read_csv(url)` loads the CSV straight from its URL into a DataFrame named `penguins`; `head()` shows the first five rows.
2. **Inspect.** `.shape` reports the size, `.info()` the column names, types and non-null counts, and `.describe()` the summary statistics of the numeric columns.
3. **Missing values.** `isna().sum()` counts the gaps per column, and the boolean filter `penguins[penguins.isna().any(axis=1)]` lists the eleven incomplete rows - the one step I found via a short AI-assistant exchange, logged in the notebook as the course AI policy requires.
4. **Select and filter.** double brackets for columns, `df[["species", "island", "body_mass_g"]]`, and a boolean mask for rows, `penguins[penguins["species"] == "Gentoo"]`, counted with `len()`.
5. **Figure.** a seaborn scatterplot of flipper length against body mass with `hue="species"`, chosen because one plot then carries two measurements and the species contrast together.

6. **Save.** the finished notebook was pushed to my dsa107-labs repository on GitHub.

4. Results

As numbers: a typical penguin in this set has a 200.9 mm flipper and a body mass of 4,201.8 g (standard deviation 802 g), and masses run from 2,700 g to 6,300 g. Sex is the patchiest column with 11 missing values (about 3.2% of rows); every measurement column is missing exactly 2. The Gentoo filter returned 124 of the 344 rows, a fraction of 0.360.

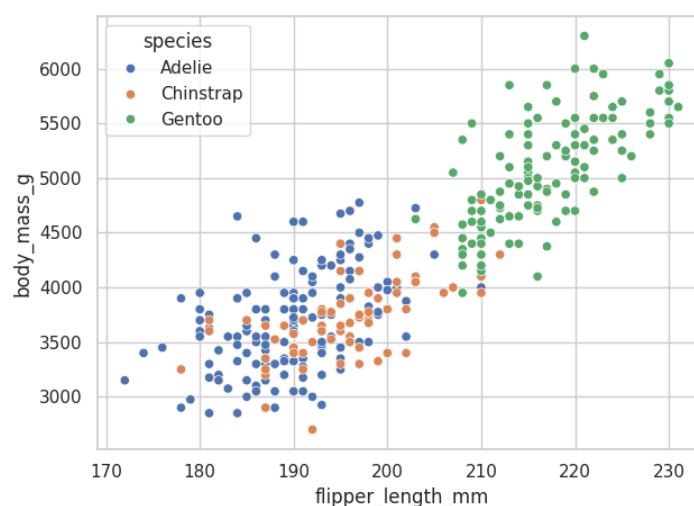


Figure 1. Flipper length versus body mass for the penguins, coloured by species (output of my notebook).

As Figure 1 shows, penguins with longer flippers are heavier. Adelie (blue) and Chinstrap (orange) points overlap in the lower-left region with flippers of roughly 172 to 210 mm, while the Gentoo points (green) occupy the upper right - flippers mostly above 210 mm and masses mostly above 4.5 kg. The upward drift is visible inside each colour as well, not only across the three species together, so the relationship holds within every species; a natural next check is whether it looks the same on all three islands.

5. Discussion

The main finding - longer flippers going with heavier bodies, within every species - is a first example of a warning the lecture course keeps repeating: a description can be true and still incomplete. Seen across the whole dataset only, part of the pattern's strength would come from the Gentoo cluster sitting apart; colouring the points by species is what separates a genuine within-group trend from a mixture of different groups.

The limits matter as much. A pattern this clean is not an explanation: nothing here says why Gentoo penguins are larger. The scatter is silently drawn from complete rows only, so Figure 1 is really 342 birds, not 344. And with 11 of 344 sexes unrecorded, any male-female comparison would first need that gap addressed. A sensible next step is a correlation computed per species and per island.

6. Reflection

What slowed me down was syntax, not concepts: I mixed up single- and double-bracket indexing more than once, and my first missing-values answer stopped at the column counts until I worked out how to list the actual rows. Writing one honest sentence under each output was slower than the code and improved my answers the most. Next time I would finish the figure in one pass - seaborn gives raw column names and no title - and paste the repository link into the notebook before closing Colab. Even so, shape, info, describe and isna gave a faster, more honest picture of the data than I expected.